NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204
Course Title: Mobile Computing Lab

Submitted By:

Team Name : CSE4204-8A-T03
Section : 8A

GitHub Repository: https://github.com/sudipto224/AI-based-Smart-Quiz-System

Submitted to:

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Designation: Assistant Professor
Department of CSE

Date of Submission: 07/09/2026

TEAM INFORMATION

Official Team Name	CSE4204-8A-T03
Section	8A
Project Title	AI-Based Smart Quiz System
GitHub Repository	https://github.com/sudipto224/AI-based-Smart-Quiz-System

Team Information:

Role	Full Name	Student ID
Team Leader + Backend	Sudipto Das	11220320830
Artificial Intelligence	Rakibur Rahman Mishan	11220320829
Database + Frontend	Aditi Roy	11220320846

Bug Tracking Method

Bugs are recorded only when an actual defect or integration issue has been observed. Severity follows the Week 10 assignment categories: Critical, High, Medium, and Low. A bug is marked Fixed only after a corrective change is implemented; a separate retest note records whether the corrected behavior has been observed or still requires a formal Week 10 regression pass.

BUG-01 :

Bug ID	BUG-01
Description	Detailed result review showed selected answers as "Not answered" even when scoring had used the student response.
Module	Quiz Attempt / Result Review
Steps to Reproduce	1. Complete a normal quiz. 2. Submit selected answers. 3. Open Detailed Review. 4. Observe that score is correct but selected answer is missing.
Expected Behavior	The result page should show the student-selected answer, correct answer, and explanation.
Actual Behavior	Selected answers were not persisted in quiz_attempts, so Detailed Review displayed "Not answered".
Severity	High
Status	Fixed
Resolution	Added an answers JSON column to quiz_attempts, updated QuizAttempt fillable/casts, saved answers during submission, and read them on the result page.
Retesting	Implementation completed; Week 10 regression re-execution is still recommended.

BUG-02 :

Bug ID	BUG-02
Description	AI question generation could produce unrelated Mobile Computing context for other selected courses.
Module	Course-aware AI Question Generator
Steps to Reproduce	1. Select a course other than Mobile Computing. 2. Enter a relevant topic. 3. Generate an AI question. 4. Observe unrelated context in generated content.
Expected Behavior	The generated MCQ should follow the selected course and teacher-entered topic.
Actual Behavior	Earlier prompt logic contained hard-coded unrelated context, causing off-topic generation.
Severity	High
Status	Fixed
Resolution	Rebuilt the AI prompt around the currently selected course and topic, removed unrelated hard-coded context, added strict JSON validation, exact option/correct-answer checks, duplicate prevention, and user-friendly error handling.
Retesting	Course-aware generation works in the current project; capture a fresh Week 10 test screenshot.

BUG-03 :

Bug ID	BUG-03
Description	The previously configured Groq model became unavailable/decommissioned, causing AI requests to fail.
Module	AI Integration
Steps to Reproduce	1. Use the older configured model. 2. Send an AI generation request. 3. Observe provider model-unavailable failure.
Expected Behavior	AI requests should use an available configured model and return structured output.
Actual Behavior	The prior Llama model selection was retired by the provider.
Severity	High
Status	Fixed
Resolution	Migrated the active AI requests to openai/gpt-oss-120b and added logging of requested/served model for verification.
Retesting	The project has verified logs showing openai/gpt-oss-120b is requested and served.

BUG-04 :

Bug ID	BUG-04
Description	Opening the teacher Live Quiz Rooms page caused a SQL error: unknown column teacher_id in quiz_sessions.
Module	Live Quiz Database / Migration
Steps to Reproduce	1. Open /teacher/live-quiz-rooms after integrating the new live-room code. 2. Laravel queries quiz_sessions by teacher_id. 3. Existing table schema lacks the new column.
Expected Behavior	Teacher live-room page should query sessions normally and render room management UI.
Actual Behavior	SQLSTATE[42S22] Column not found: 1054 Unknown column teacher_id in where clause.
Severity	Critical
Status	Fixed
Resolution	Created a safe repair_live_quiz_tables migration that conditionally creates/adds the required live-session, pivot, participant, and quiz_attempt relationship columns without using migrate:fresh or deleting existing data.
Retesting	After migration, /teacher/live-quiz-rooms loaded successfully and a new room was created.

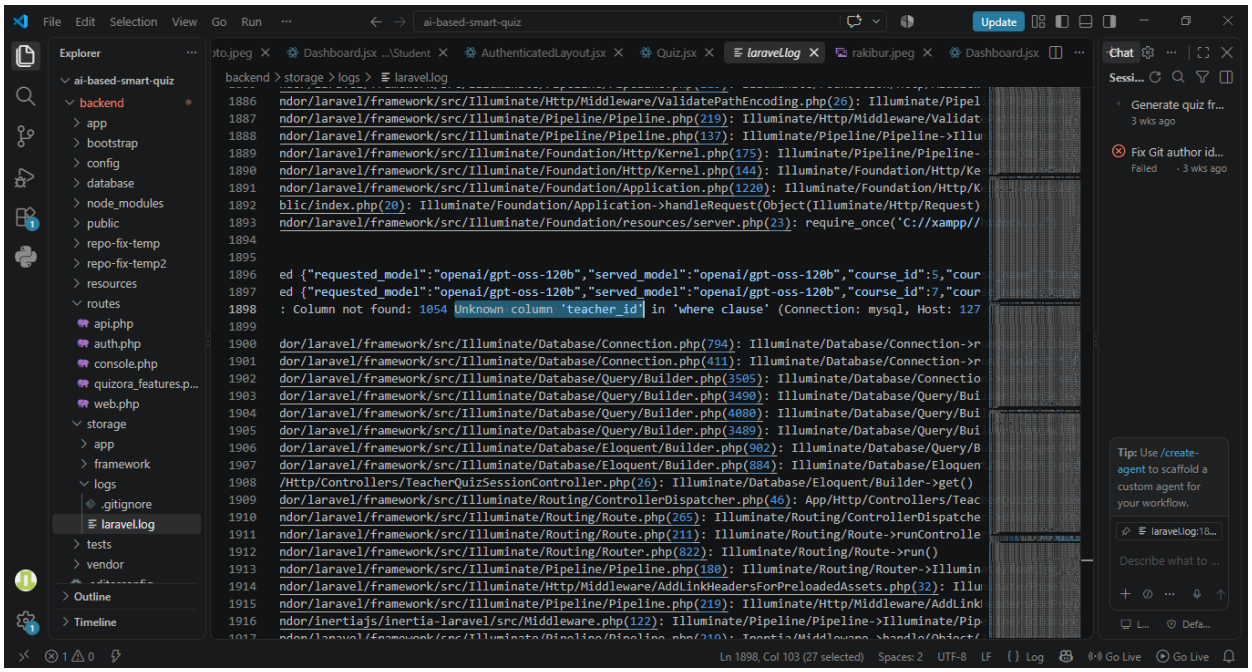


Figure: BUG-04 before fix: SQL error showing unknown `teacher_id` column in the `quiz_sessions` table.

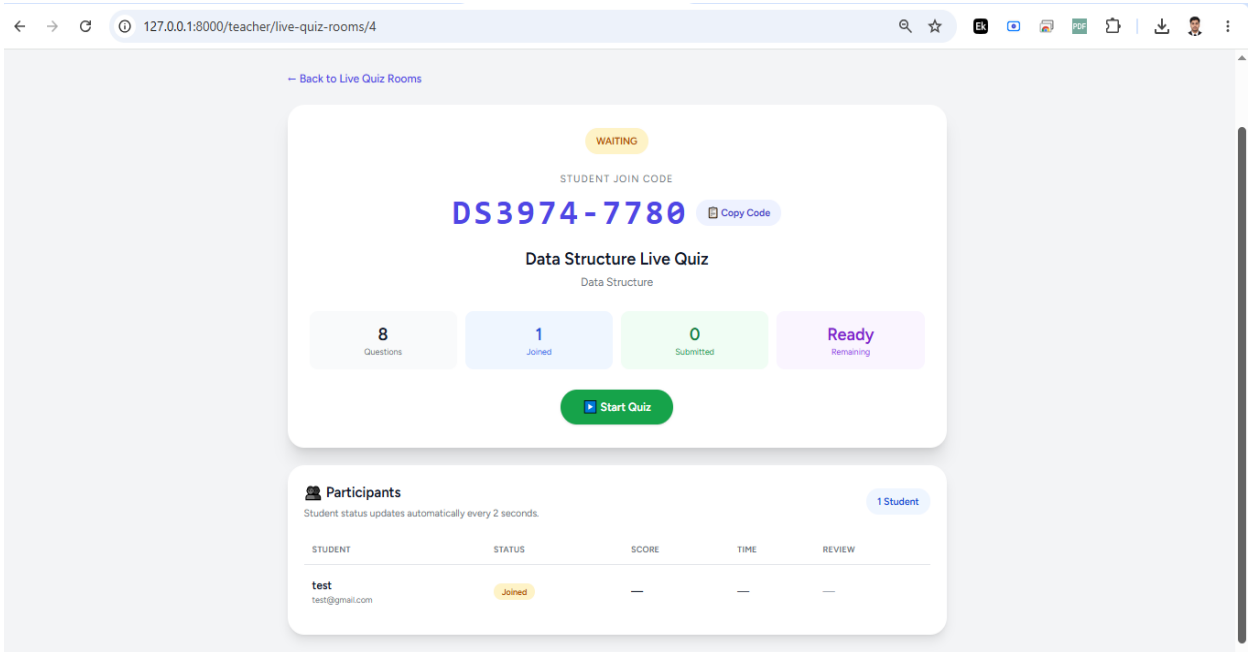


Figure: BUG-04 After Fix: Live Quiz Rooms page successfully loaded after resolving the database column error.

BUG-05 :

Bug ID	BUG-05
Description	Teacher “Open Room” navigated to /teacher/live-quiz-rooms/1 but displayed a blank white page.
Module	Teacher Live Quiz Room UI / Inertia
Steps to Reproduce	1. Create a live room. 2. Click Open Room. 3. Browser navigates to the room URL. 4. Page remains blank. 5. Console shows Page not found: ./Pages/Teacher/QuizRooms/Show.jsx.
Expected Behavior	The teacher should see room code, status, question count, participants, and Start/End controls.
Actual Behavior	The Inertia page component Show.jsx was missing or not located at the exact expected path.
Severity	High
Status	Fixed
Resolution	Created/moved the full Teacher/QuizRooms/Show.jsx component to resources/js/Pages/Teacher/QuizRooms/Show.jsx and rebuilt/cleared caches.
Retesting	Room detail page rendered correctly with WAITING status, DS-3974 code, 5 questions, 1 joined participant, and Start Quiz button.

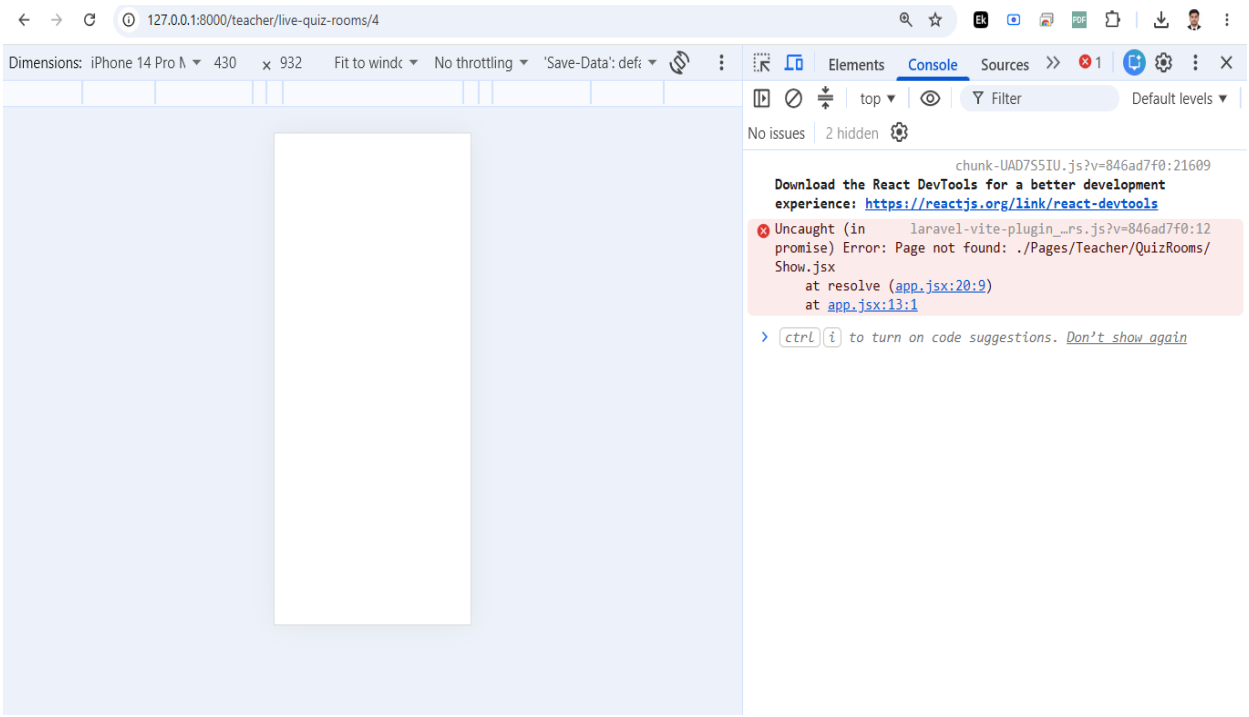


Figure: BUG-05 Before Fix: Blank teacher live quiz room page caused by missing Show.jsx component.

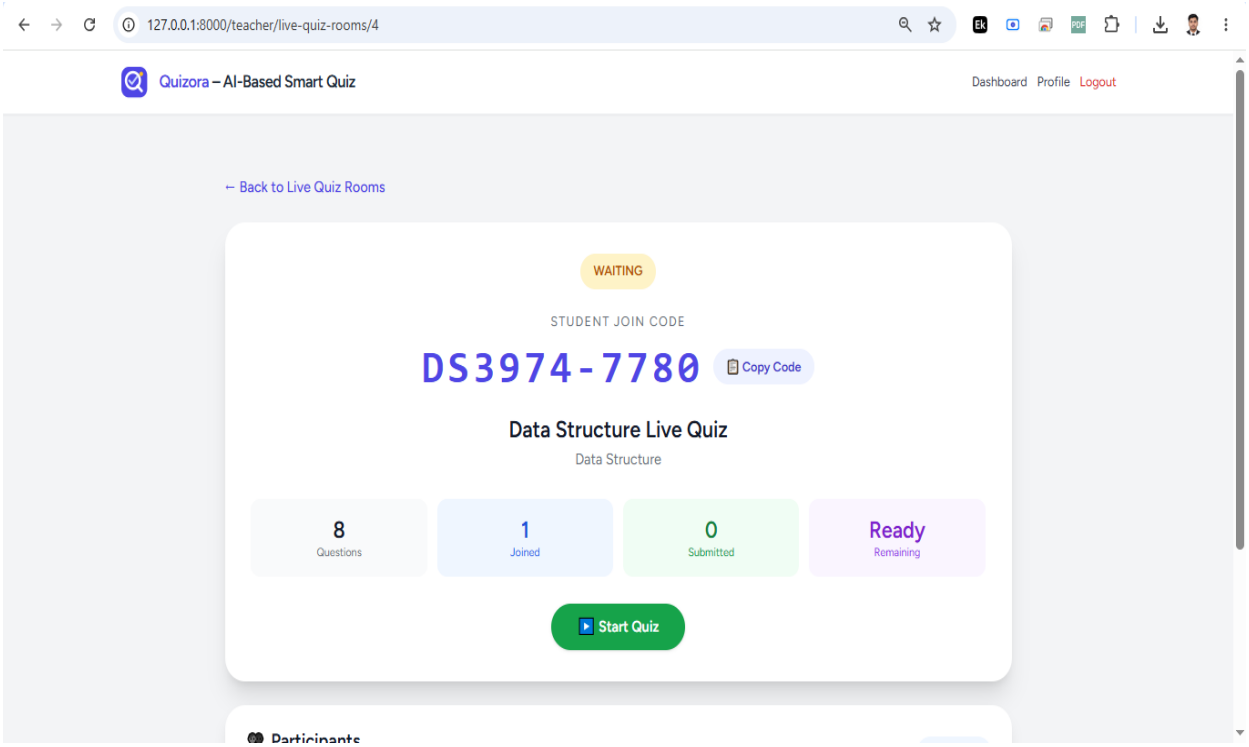


Figure: BUG-05 After Fix: Teacher room details page successfully loaded, showing join code DS3974-7780 and the Start Quiz button.

BUG-06 :

Bug ID	BUG-06
Description	After the teacher started the live quiz, the student was redirected to /student/live-quiz/1/quiz but the page was blank.
Module	Student Live Quiz UI / Inertia
Steps to Reproduce	1. Student joins room and waits. 2. Teacher clicks Start Quiz. 3. Student redirects to /student/live-quiz/1/quiz. 4. Page is blank. 5. Console shows Page not found: ./Pages/Student/QuizRooms/Quiz.jsx.
Expected Behavior	Student should see live questions, shared timer, answer options, and Submit Quiz button.
Actual Behavior	Quiz.jsx existed outside the QuizRooms folder, so Inertia could not resolve the requested page component.
Severity	High
Status	Fixed
Resolution	Moved Quiz.jsx to resources/js/Pages/Student/QuizRooms/Quiz.jsx and ensured Result.jsx was also created in the same QuizRooms folder.
Retesting	The component path was corrected; final Week 10 screenshot of a full live quiz and result should be captured.

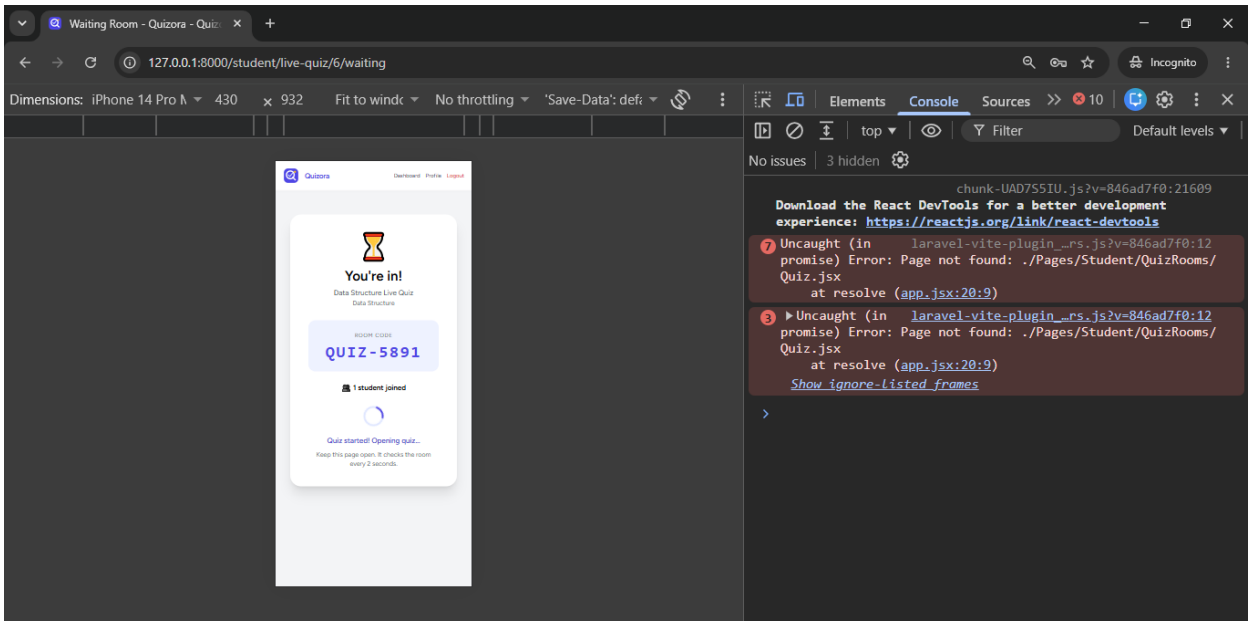


Figure: BUG-06 Before Fix – Blank student quiz page due to missing Quiz.jsx component path error.

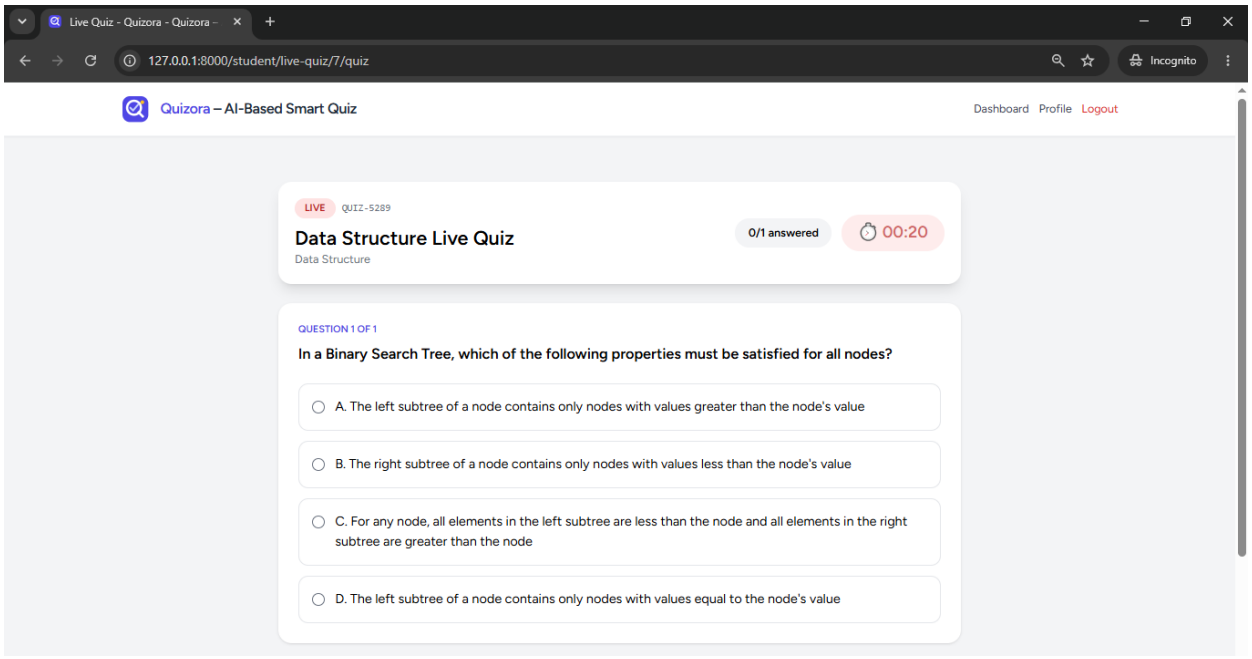


Figure: BUG-06 After Fix – Student live quiz page successfully loaded with questions, answer options, and countdown timer.

BUG-07 :

Bug ID	BUG-07
Description	Frontend changes sometimes did not appear immediately after editing React files.
Module	Frontend Build / Development Environment
Steps to Reproduce	1. Edit a React component. 2. Refresh browser. 3. Old compiled UI may still appear.
Expected Behavior	Saved frontend changes should be reflected in the active development build.
Actual Behavior	Browser/Vite/build caches occasionally served older compiled assets.
Severity	Low
Status	Fixed / Workaround
Resolution	Use npm run dev during development or rebuild with npm run build, then php artisan optimize:clear and Ctrl+Shift+R hard refresh when necessary.
Retesting	New dashboard/live-room components displayed after correct file placement and rebuild/refresh.

Bug Summary :

Bug ID	Module	Severity	Status	Retest State
BUG-01	Quiz Attempt / Result Review	High	Fixed	Formal Week 10 retest recommended
BUG-02	Course-aware AI Question Generator	High	Fixed	Formal Week 10 retest recommended
BUG-03	AI Integration	High	Fixed	Verified / observed
BUG-04	Live Quiz Database / Migration	Critical	Fixed	Verified / observed
BUG-05	Teacher Live Quiz Room UI / Inertia	High	Fixed	Verified / observed
BUG-06	Student Live Quiz UI / Inertia	High	Fixed	Formal Week 10 retest recommended
BUG-07	Frontend Build / Development Environment	Low	Fixed / Workaround	Verified / observed

Remaining Bug-Tracking Actions

- Any Pending test case that fails during Week 10 must receive a new Bug ID and be added to this report.
- After every fix, repeat the original reproduction steps and record the new actual behavior.
- After major fixes, run regression tests on authentication, database, normal quiz, AI, and Live Quiz modules.
- Keep before-fix and after-fix screenshots for the most important defects.
- Use meaningful GitHub commit messages that describe the specific bug or test improvement.